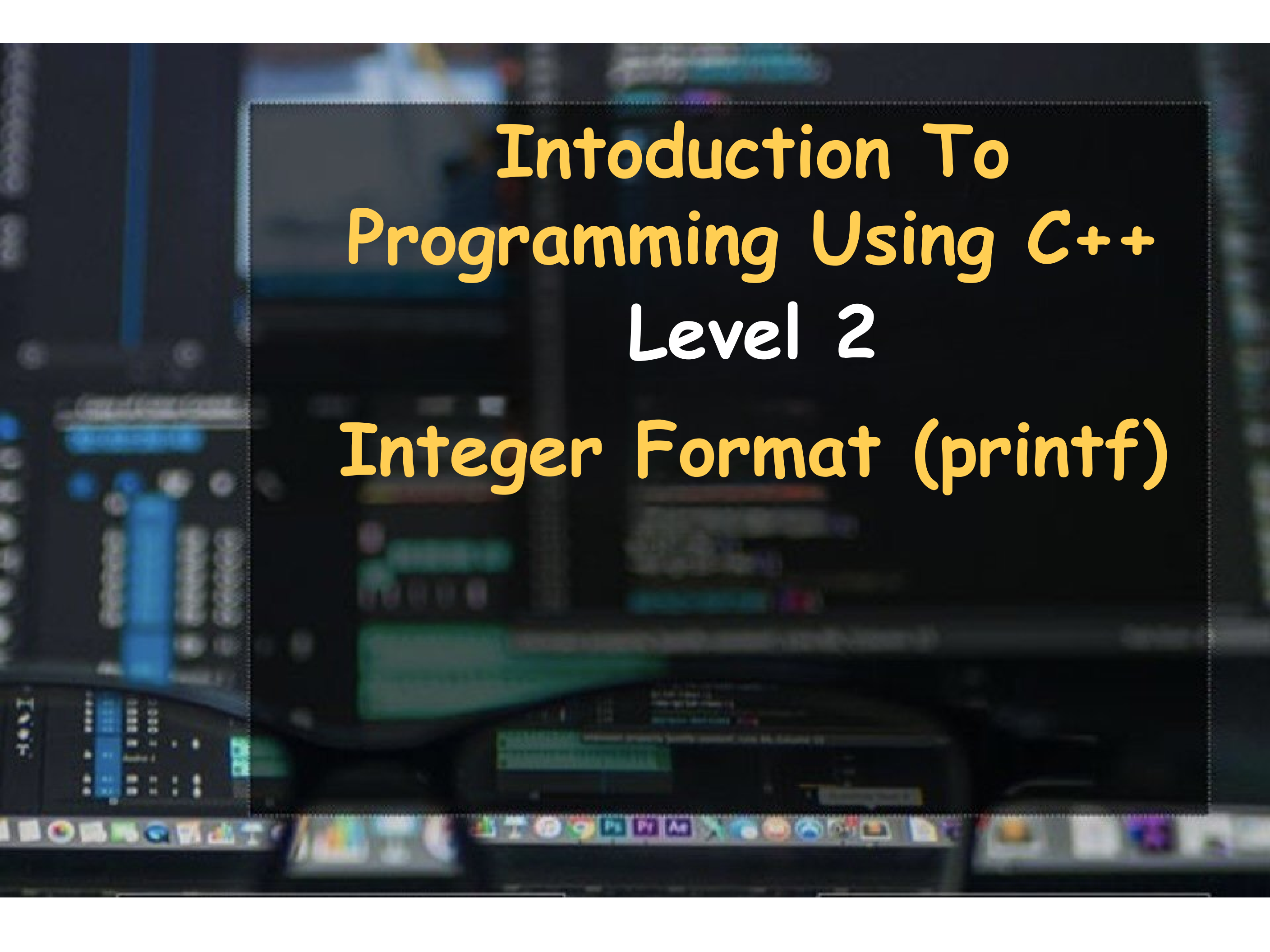
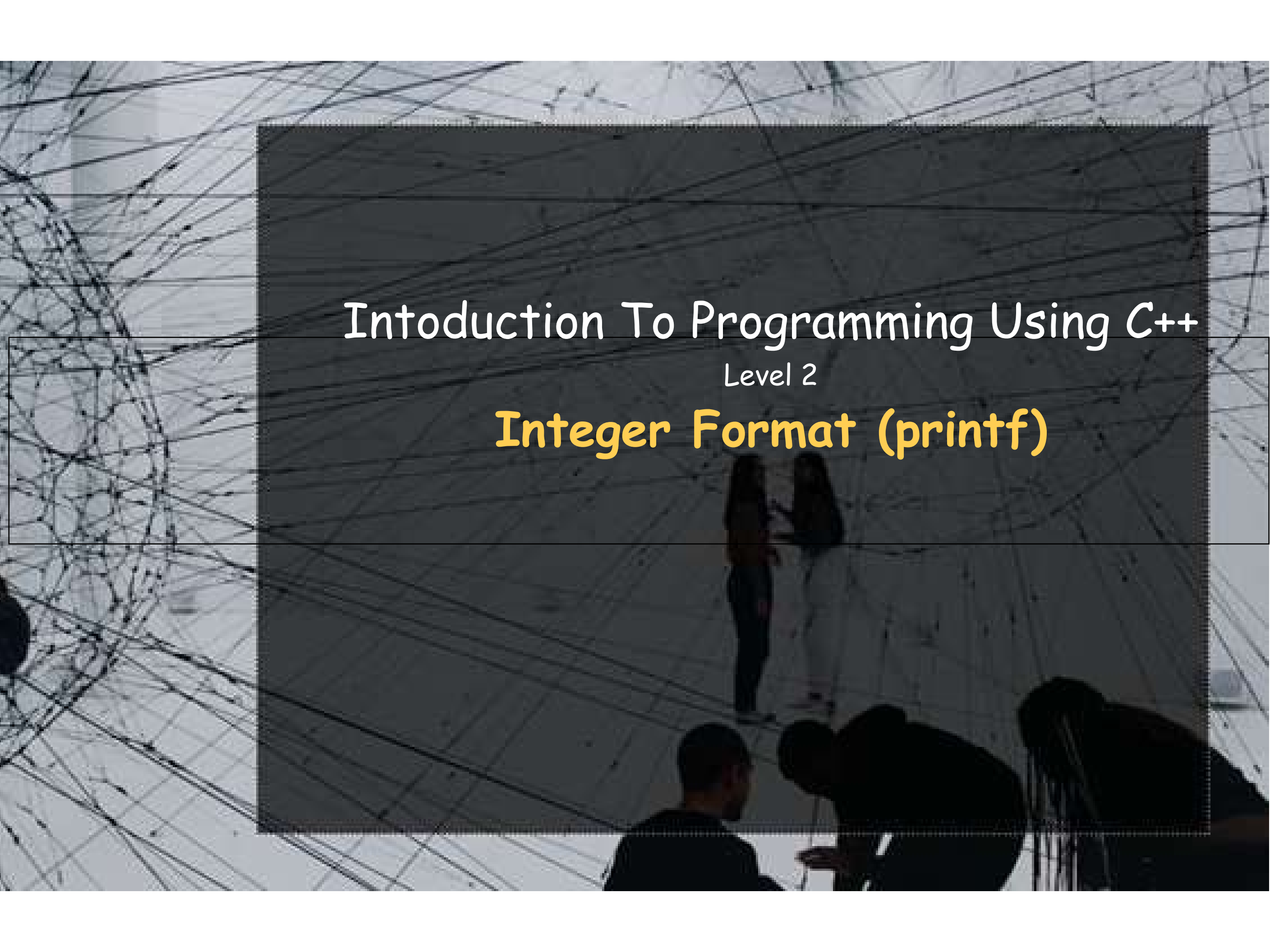


The background of the slide is a blurred image of a computer monitor. On the left side, a calculator application is visible, showing a numeric keypad and some text. The main area of the monitor displays what appears to be a code editor with lines of text in various colors (green, blue, white) on a dark background, typical of a programming IDE. At the bottom of the monitor, a taskbar with several application icons is visible.

Introduction To Programming Using C++ Level 2 Integer Format (printf)



Introduction To Programming Using C++

Level 2

Integer Format (printf)

Definition

■ `printf()` is a function from the C Standard Library that can also be used in C++ to print formatted output.

■ It uses Format Specifiers inside the text to determine how values should be displayed.

■ Common Format Specifiers include:

▨ `%d` → prints an Integer (`int`)

▨ `%f` → prints a Floating-Point value

▨ `%s` → prints a String

Explanation

◆ The basic syntax of `printf()` is:

`printf("Text with Format Specifiers", values);`

◆ The Format Specifier tells `printf()` what Type of value should be formatted and printed.

◆ In this lesson, we focus mainly on Integer Formatting using `%d`.

Procedures

1. `%d``

- Used to print an Integer value.

```
printf("%d", Number);
```

2. `%0*d``

- Used to specify the Width of the output and fill empty positions with zeros.

- The number before the Integer Variable specifies the required Width.

```
printf("%0*d", 3, Page);
```

- If `Page = 1``, the output is:

001

3. Multiple `%d`` Specifiers

- Multiple Integer values can be printed in the same statement.

- You can also perform calculations directly inside the arguments.

```
printf("The Result of %d + %d = %d", Number1, Number2, Number1 + Number2);
```

💡 Example

```
int main()
{
    int Page = 1, TotalPages = 10;
    // Print an Integer Variable
    printf("The page number = %d\n", Page);
    // Print multiple Integer Variables
    printf("You are in Page %d of %d\n", Page, TotalPages);
    // Width specification
    printf("The page number = %0*d\n", 2, Page);
    printf("The page number = %0*d\n", 3, Page);
    printf("The page number = %0*d\n", 4, Page);
    printf("The page number = %0*d\n", 5, Page);
    int Number1 = 20, Number2 = 30;
    // Print a calculation
    printf(
        "The Result of %d + %d = %d\n",
        Number1,
        Number2,
        Number1 + Number2
    );
    return 0;
}
```

💡 Example

➡ Result

The page number = 1

You are in Page 1 of 10

The page number = 01

The page number = 001

The page number = 0001

The page number = 00001

The Result of $20 + 30 = 50$

Note

- ◆ `printf()` uses the Format Specifiers to determine how the provided arguments should be formatted.
- ◆ The number of Format Specifiers should correspond to the values provided after the Format String.
- ◆ `\n` moves the cursor to a new line.

⊘ Warning

- ◆ The Format Specifier must match the Type of the corresponding argument.
- ◆ For example, `%d` is intended for an `int`. Using an incompatible Format Specifier can lead to incorrect output or undefined behavior.



Comparison

▨ ``%d``

- ✓ Used to print an Integer.
- ✓ Example: ``printf("%d", 25);``
- ✓ Does not add leading zeros by itself.

▤ ``%0*d``

- ✓ Used when a specific Width with leading zeros is required.
- ✓ Example: ``printf("%0*d", 4, 25);``
- ✓ Result: ``0025``

✦ Summary

- ✓ `printf()` is used to produce formatted output.
- ✓ `%d` is used to print Integer values.
- ✓ `%0*d` allows you to specify the Width and fill empty positions with zeros.
- ✓ Multiple `%d` Specifiers can be used in the same `printf()` statement.
- ✓ Calculations can be performed directly inside the arguments.
- ✓ `printf()` is commonly used in C and is also available in C++ through `<cstdio>`.

Thank you 😊